Bankers Algorithm

Input

user07@localhost:~/bankers — /usr/bin/vim banker.c

```c
#include<stdio.h>
int main()
{
int n,r,i,j,k;
printf("Enter no of processes");
scanf("%d",&n);
printf("Enter no of resource types");
scanf("%d",&r);
int alloc[n][r],max[n][r],need[n][r],avail[r];
int finish[n],safe_sequence[n];
int count=0,safefound;
printf("Enter allocation matrix:\n");
for(i=0;i<n;i++)
{
for(j=0;j<r;j++){
scanf("%d",&alloc[i][j]);
}
}
printf("Enter max matrix:\n");
for(i=0;i<n;i++){
for(j=0;j<r;j++){
scanf("%d",&max[i][j]);
}
}
printf("Enter available resources:\n");
for(i=0;i<r;i++){
scanf("%d",&avail[i]);
}
for(i=0;i<n;i++){
finish[i]=0;
-- INSERT --                                                    1,2           Top
```

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```c
}
for(i=0;i<n;i++){
for(j=0;j<r;j++){
need[i][j]=max[i][j]-alloc[i][j];
}
}
while(count<n){
safefound=0;
for(i=0;i<n;i++){
if(finish[i]==0)
{
int safe = 1;
for(j=0;j<r;j++)
{
if(need[i][j]>avail[j])
{
safe=0;
break;
}
}
if(safe)
{
for(k=0;k<r;k++)
{
avail[k]=avail[k]+alloc[i][k];
}
safe_sequence[count]=i;
finish[i]=1;
safefound=1;
count ++;
-- INSERT --                                                    55,2          68%
```

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```
if(need[i][j]>avail[j])
{
safe=0;
break;
}
}
if(safe)
{
for(k=0;k<r;k++)
{
avail[k]=avail[k]+alloc[i][k];
}
safe_sequence[count]=i;
finish[i]=1;
safefound=1;
count ++;
}
}
}
if(!safefound)
{
printf("System is in unsafe state! may occur the deadlock \n");
return 0;}
}
printf("safe sequence is: ");
for(i=0;i<n;i++){
printf("P%d ",safe_sequence[i]);
}
return 0;
}
-- INSERT --                                                    74,2          Bot
```

Output

user07@localhost:~/bankers

```
      |     ^
banker.c:74: error: expected '{' at end of input
[user07@jannani bankers]$ vi banker.c
[user07@jannani bankers]$ gcc banker.c
banker.c: In function 'main':
banker.c:39:7: error: 'o' undeclared (first use in this function)
   39 | for(i=o;i<n;i++){
      |       ^
banker.c:39:7: note: each undeclared identifier is reported only once for each function it appears in
[user07@jannani bankers]$ vi banker.c
[user07@jannani bankers]$ gcc banker.c
[user07@jannani bankers]$ ./a.out
Enter no of processes5
Enter no of resource types4
Enter allocation matrix:
2 0 0 1
3 1 2 1
2 1 0 3
1 3 1 2
1 4 3 2
Enter max matrix:
4 2 1 2
5 2 5 2
2 3 1 6
1 4 2 4
3 6 6 5
Enter available resources:
3 3 2 1
safe sequence is: P0 P3 P4 P1 P2 [user07@jannani bankers]$ P0 P3 P4 P1 P2
```